SHINAS SAMOON

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PROFESSIONAL SUMMARY

B.Tech ECE graduate with hands-on experience in Python, AI/ML, web scraping, backend, and full-stack development. Skilled in Flask, Django, FastAPI, React, TypeScript, REST APIs, MongoDB, PostgreSQL, Redis, Playwright, Selenium, BeautifulSoup, LangChain, RAG, and Generative AI. Experienced in building and deploying real-world applications, automation tools, data pipelines, and AI-powered solutions.

EDUCATION

Bachelor of Technology in Electronics and Communication Engineering 2022 – 2026
L.B.S College of Engineering Kasaragod CGPA: 7.04

Higher Secondary Education 2019 – 2021
Rahmania HSS For Handicapped, Kozhikode 95%

High School Education 2018 – 2019
P.T.M.H.S Kodiyathur, Kozhikode 90%

PROJECTS

UWB Based Indoor Activity Monitoring for Elderly Care

- Developed a real-time indoor monitoring system using UWB transceivers, ESP32, and sensors.
- Built Python-based machine learning models for activity classification and fall detection.
- Designed data preprocessing and feature extraction pipelines.
- Integrated Telegram APIs for automated emergency notifications.

PDF RAG Chatbot | Python, LangChain, FAISS, Gemini, Streamlit

- Built a Retrieval-Augmented Generation (RAG) application for intelligent PDF document querying.
- Processed PDFs into vector embeddings using LangChain and a vector database.
- Retrieved relevant document chunks through semantic search before generating answers with an LLM.
- Improved response accuracy by grounding AI responses in source documents.

Distributed Media Scraping Link Resolution Engine — Python, MongoDB, Requests, RapidFuzz, Proxy Rotation

- Engineered an automated web scraping pipeline extracting media catalog data and stream links across 15+ high-traffic streaming sources with custom DOM/Regex parsers.
- Implemented an anti-bot evasion system featuring rotating proxy pools, custom headers manipulation, and session handling to bypass rate limits and IP bans with 99% scraping uptime.
- Built a multi-service bypass and token-resolution engine decoding obfuscated URLs and multi-step redirect chains into final endpoints.
- Designed an intelligent metadata enrichment layer using RapidFuzz string matching and TMDb API integration to cleanse, standardize, and associate extracted titles with IMDb/TMDb IDs.
- Maintained a high-throughput MongoDB caching architecture for link deduplication, incremental crawling, and state persistence.

Malayalam Flash News — AI-Powered Regional News Aggregator — Next.js, TypeScript, FastAPI, Python, Google Gemini AI, Tailwind CSS

- Engineered an automated end-to-end news summarization platform aggregating real-time regional news across 5+ major publishers.
- Reverse-engineered Google News' batch URL resolution protocol (batchexecute) to extract and resolve canonical publisher URLs from encoded RSS feeds.
- Integrated Google Gemini LLM with custom Malayalam prompt engineering and Google Search Grounding tool fallback, generating 50–80 word clickbait-free news summaries in structured JSON.
- Implemented a multi-tier hashing and TTL-based caching layer for both scraped content and LLM outputs, cutting Gemini API token consumption and reducing response latency by 80%.
- Built a responsive, dual-theme frontend (Neo-Brutalist Swiss International Typographic styles) featuring CSS scroll-snapping, live search, publisher filtering, and Web Share API integration.

INTERNSHIPS & CERTIFICATIONS

- **Machine Learning & AI Intern, ICT Academy of Kerala** – Worked on data preprocessing, feature engineering, model training, and evaluation using Python.
- Completed **Python Training Program organized by L.B.S College of Engineering Kasaragod**.
- Published and presented **"UWB Based Indoor Activity Monitoring for Elderly Care"** at ICSTS 2025.
- LinkedIn Learning: Machine Learning with Python – k-Means Clustering.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C

Frameworks & Technologies: Flask, React, REST APIs, Streamlit, Next.js, Tailwind, git,playwright, pytest, web scraping

Databases: MongoDB, SQL, PostgreSQL, Supabase, Neon

Tools: Git, GitHub, Postman, Linux, MATLAB, Google Colab, Burp Suite

Software Engineering: Object-Oriented Programming (OOP), Data Structures & Algorithms, Debugging, API Development, Version Control, Testing, SDLC

AI & ML: NumPy, Scikit-Learn, RAG, MCP, AI Agent, Claude, Vector Databases, Prompt Engineering

Hardware & IoT: ESP32, ESP8266, Arduino IDE, Embedded Systems

Soft Skills: Problem Solving, Team Collaboration, Communication, Leadership